R1 # show ip nat statistics
Total active translations: 2 (2 static, 0 dynamic; 0 extended)
peak-translation = 4 (highest translation in NAT table)

Day-45

June-14.

NAT (PART-II) - Dynamic NAT

More about Static NAT

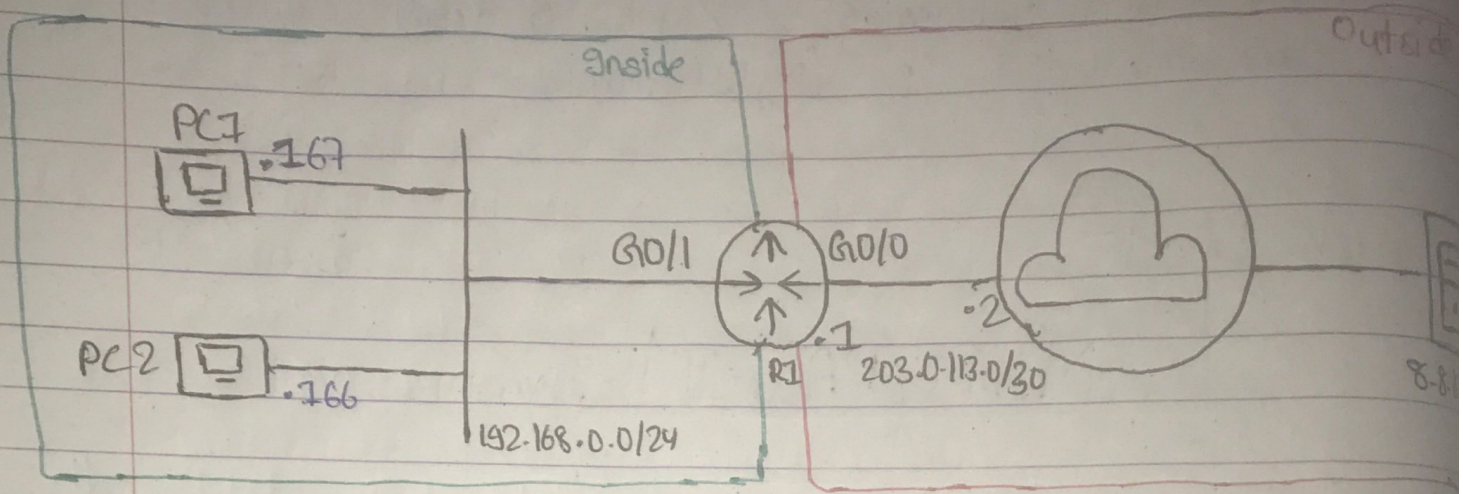
- Static NAT involves statistically configuring one-to-one mappings of private IP addresses to public IP addresses.
- When traffic from internal host is sent to the outside network, the router will translate the source address.
- However, this one-to-one mapping also allows external hosts to access the internal host via the inside global address.

Dynamic NAT

- In dynamic NAT, the router dynamically maps 'inside local add.' to 'inside global add.' as needed. and clears the mapping when its no longer needed.
- An ACL is used to identify which traffic should be translated.
  - ↳ If the src IP is permitted by the ACL, the src IP will be translated.
  - ↳ If the source IP is denied by ACL, the src IP will Not be translated. (* the traffic won't be dropped)

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* Network diagram is as of part I.



- A NAT pool is used to define the available 'inside global addresses'.

On R1:

ACL 1: permit 192.168.0.0/24
deny any

POOL 1: 100.0.0.1 to 100.0.0.10

If a packet with src IP permitted by ACL 1 arrives, translates the src IP to an address from POOL1

It same as static, but now what happens is when packet from 192.168.0.167 arrives on R1, instead of specifying mapping one-to-one, it happens automatically.

- Although they are dynamically assigned, the mappings are still one-to-one (one inside local IP address per inside global IP address).

- If there aren't enough 'inside global IP address available (= all are currently being used), it is called 'NAT pool exhaustion'
  - ↳ If a packet from another 'inside host arrives and needs NAT but there are no available addresses, the router will drop the packet.
  - ↳ The host'll be unable to access outside networks until one of the 'inside global IP add. becomes available.
  - ↳ Dynamic NAT entries will timeout automatically, if not used, or we can clear them manually.

To clear manually: clear ip nat translations

Dynamic NAT Configuration.

R1(config)# int g0/1
R1(config-if)# ip nat inside.
} Define the 'inside' interface(s) connected to internal network.

R1(config-if)# int g0/0
R1(config-if)# ip nat outside
} Define the 'outside' interface(s) connected to external network.

R1(config)# access-list 1 permit 192.168.0.0 0.0.0.255
Define the traffic that should be translated.
* Traffic permitted by this ACL will be translated.

R1(config)# ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24.
(100.0.0.0 = first-ip-in-range; 100.0.0.255 = last-ip-in-range)

Define the pool of inside global IP add.
* instead of prefix-length 24, we can use netmask 255.255.255.0

R1(config)# ip nat inside source list 1 pool POOL1.
Configure dynamic NAT by mapping the ACL to the pool.

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** Send some pings from both PCs to outside add. & check translation table to confirm.
R1 # show ip nat translations.
R1 # show ip nat statistics

↳ The original dynamic mappings themselves have a default timeout value of 24 hours & each-time translation is made timer resets. We can change default timing frm 24 hrs (but not needed to learn it for CCNA)

PAT (Port Address Translation) / NAT Overload.

- PAT (aka NAT overload) translates both IP add. & the port no (if necessary).

- By using a unique port no. for each communication (flow between internal & external hosts, a single public IP add. can be used by diffnt internal hosts (port no. are 16 bits = over 65k available port numbers)

- The router will keep track of which inside local address is using which inside global address and port.

Example.

PC1 sends DNS mssg to 8.8.8.8

Src	Dst
192.168.0.167: 54321	8.8.8.8:53

PC2 also does same & ~~mistakenly~~ coincidentally port no. of src is same

Src	Dst
~~192~~ 192.168.0.168: 54321	~~192~~ 8.8.8.8:53

Now R1 changes the mssg from PC1 as below

Src: ~~192.168~~ 100.0.0.1:54321 | Dst: 8.8.8.8:53

Now for R2, it does as below

Src: 100.0.0.1:54322 | Dst: 8.8.8.8:53

Now when SRV-8.8.8.8 sends reply, R1 ~~has~~ sends repl that reply to PC1 & PC2 based on port numbers

Continue....

- Because many inside hosts can share a public IP, PAT is very useful for preserving public IP add, & is used in networks all over the world.

PAT Configuration - (pool method)

R1 (config) # int g0/1
R1 (config-if) # ip nat ~~outside~~ inside } Define inside int connected to network

R1 (config) # int g0/0
R1 (config-if) # ip nat outside } Define outside int. connected to network
R1 (") # exit

R1 (config) # access-list 1 permit 192.168.0.0 0.0.0.255
- Defines traffic that should be translated (ie permitted by ACL)

R1 (config) # ip nat pool POOL1 100.0.0.0 100.0.0.3 ~~prefix~~ prefix-length 24
(Defines pool of inside global IP addresses)

R1 (config) # ip nat inside source list 1 pool POOL1 overload
Configured PAT by mapping the ACL to the pool & using the overload keyword at end

~~Ufo~~

NOB: PAT - many-to-one mapping

PAT Configuration (Interface)

- Configures the router to use its own public IP add. when translating the src IP of packets.

R1(config)# int g0/1
R1(config-if)# ip nat inside

R1(config-if)# int g0/0
R1(config-if)# ip nat outside

R1(config)# access-list 1 permit 192.168.0.0 0.0.0.255

R1(config)# ip nat inside source list 1 interface g0/0 overload